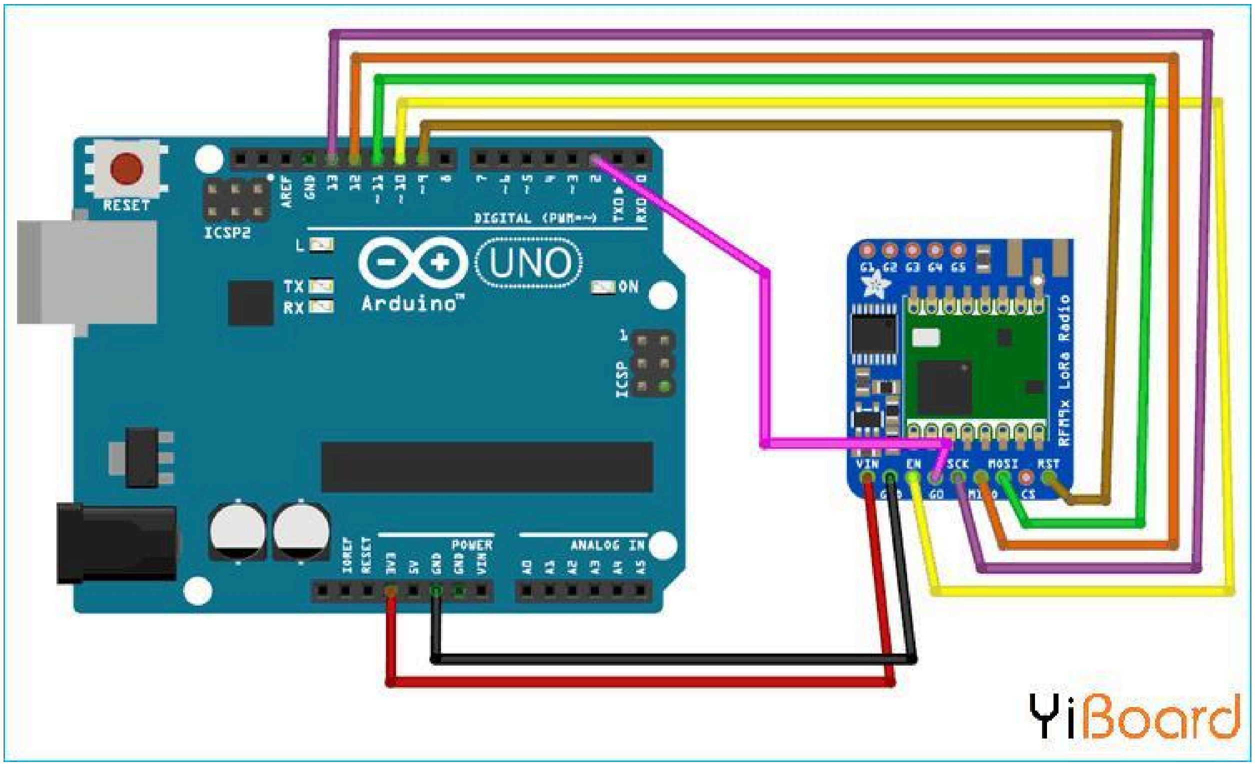
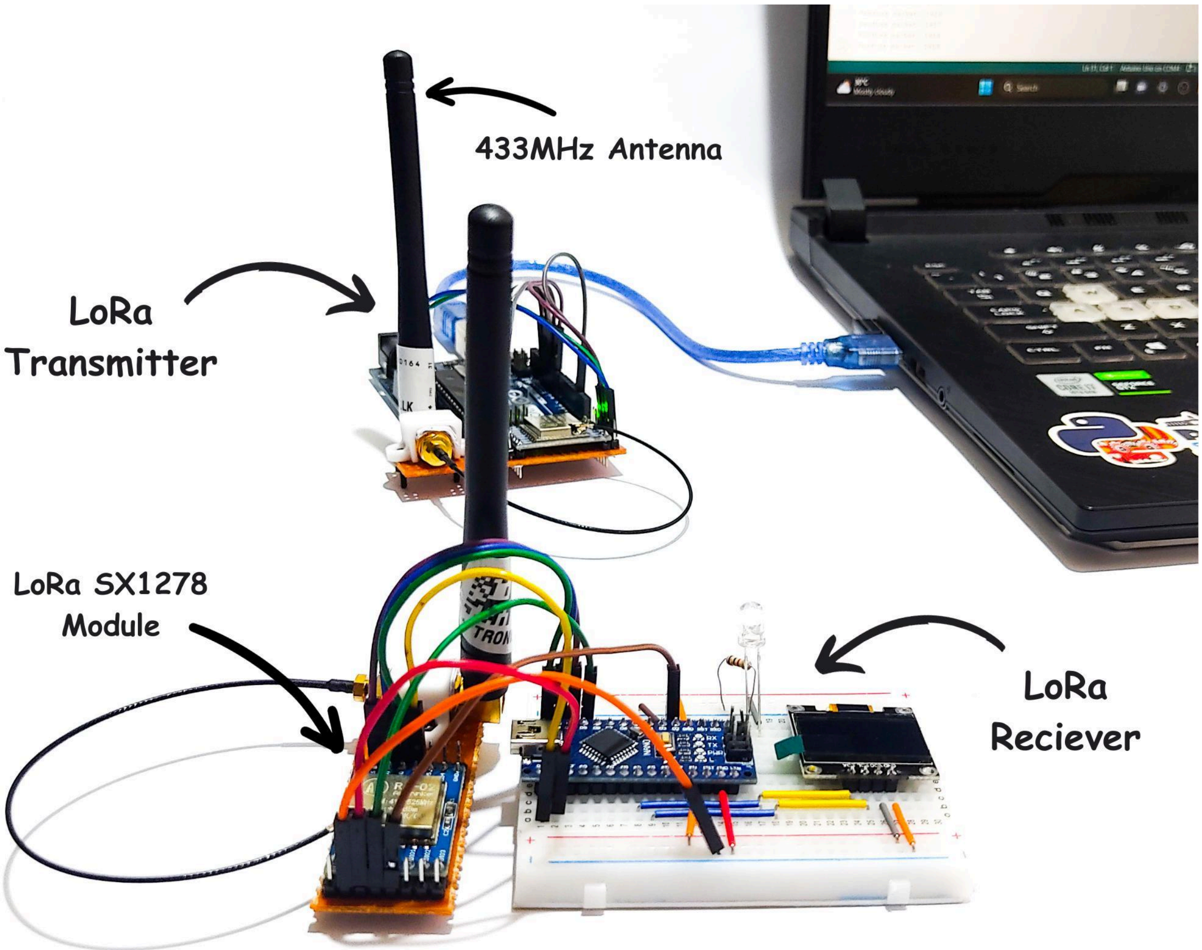
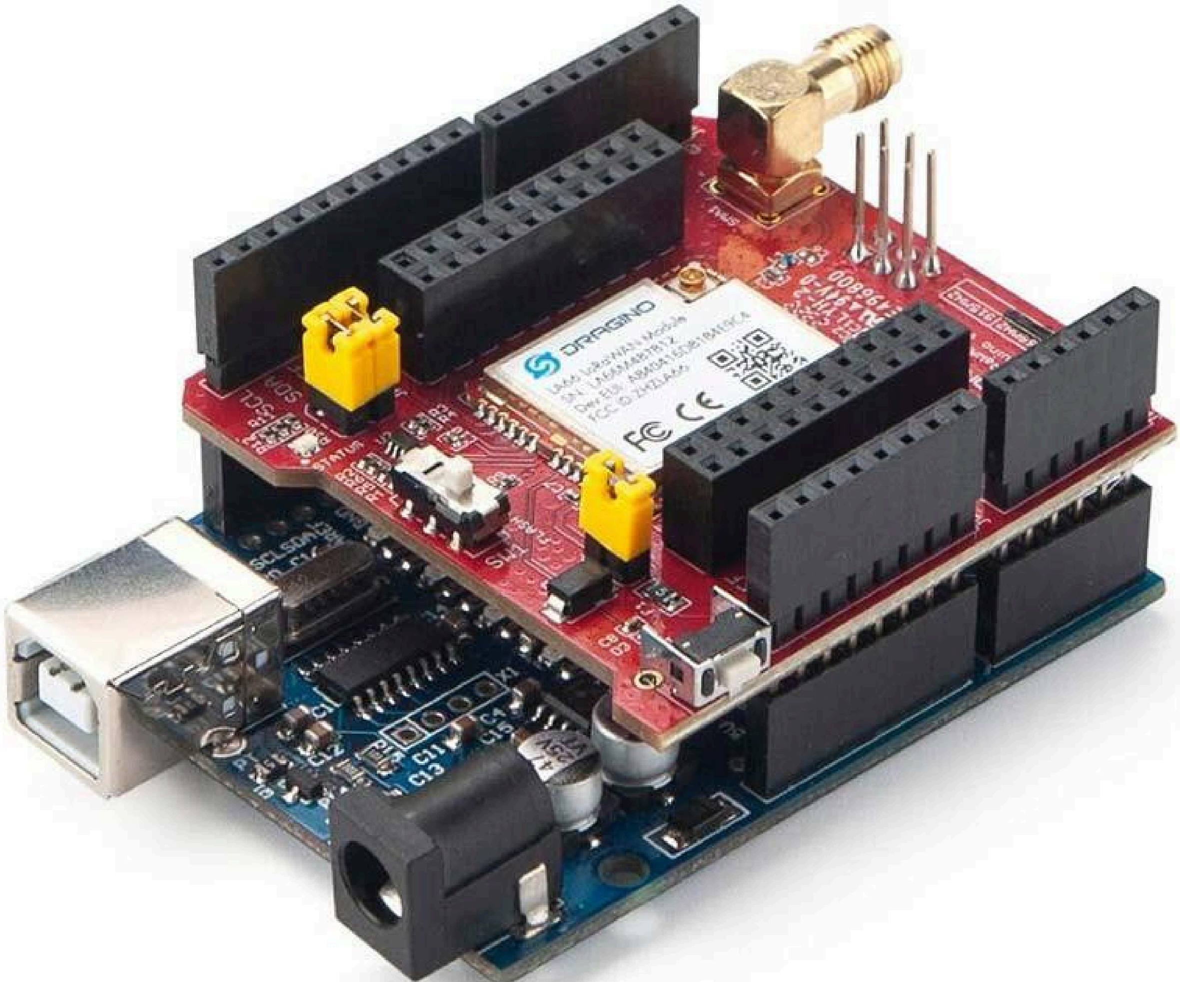
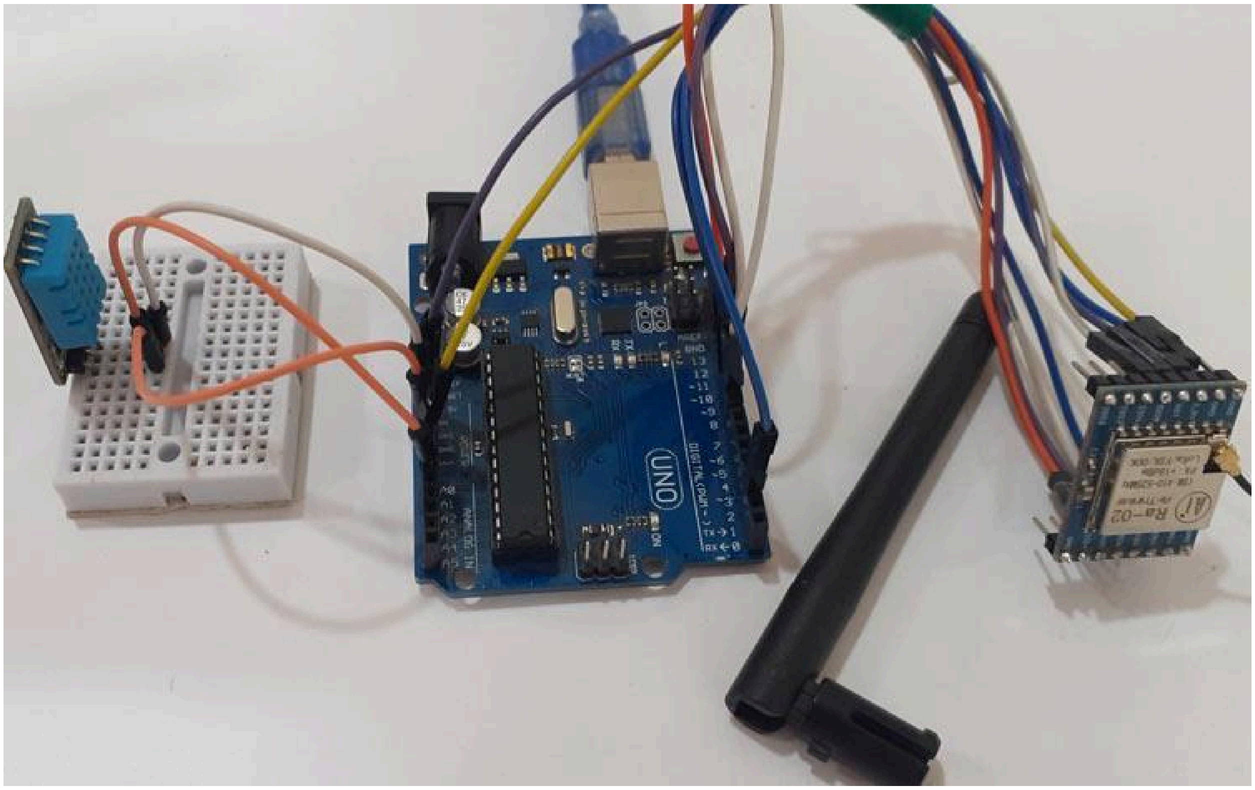
Yes 👍
You can make a simple **LoRa message system using Arduino + SX1262/SX1278** and easily get **2 km distance**.

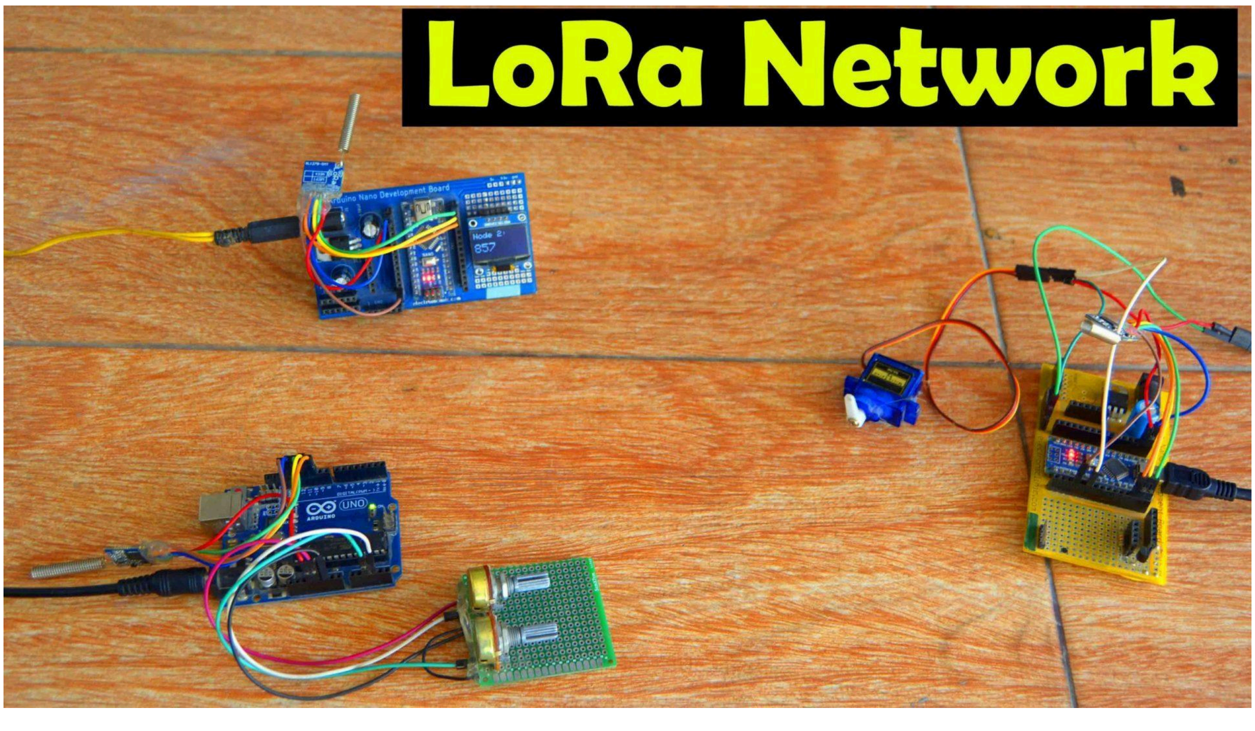
Required Components




433MHz Antenna
LoRa
Transmitter
LoRa SX1278
Module
LoRa
Reciever


DRAGINO




LoRa Network

- 2 × Arduino Uno / Nano
- 2 × LoRa modules (SX1278 / SX1262)
- 2 × antennas
- jumper wires

Wiring (SX1278 → Arduino UNO)

LoRa	Arduino UNO
VCC	3.3V
GND	GND
SCK	D13
MISO	D12
MOSI	D11

NSS	D10
RST	D9
DIO0	D2

⚠️ Important:
Most LoRa modules use **3.3V**.

Install Arduino Library

Open Arduino IDE:

Sketch
→ Include Library
→ Manage Libraries

Search:

LoRa by Sandeep Mistry

Install it.

Sender Code

```
#include <SPI.h>
#include <LoRa.h>

void setup() {
  Serial.begin(9600);

  LoRa.setPins(10, 9, 2);

  if (!LoRa.begin(865E6)) {
    Serial.println("LoRa Failed!");
    while (1);
  }
```

```
  Serial.println("LoRa Sender Started");
}

void loop() {

  LoRa.beginPacket();
  LoRa.print("Hello from Arduino");
  LoRa.endPacket();

  Serial.println("Message Sent");

  delay(2000);
}
```

Receiver Code

```
#include <SPI.h>
#include <LoRa.h>

void setup() {
  Serial.begin(9600);

  LoRa.setPins(10, 9, 2);

  if (!LoRa.begin(865E6)) {
    Serial.println("LoRa Failed!");
    while (1);
  }

  Serial.println("LoRa Receiver Started");
}

void loop() {

  int packetSize = LoRa.parsePacket();

  if (packetSize) {

    Serial.print("Received: ");
```

```
    while (LoRa.available()) {
      Serial.print((char)LoRa.read());
    }

    Serial.println();
  }
}
```

Upload Steps

Sender Arduino

Upload sender code.

Receiver Arduino

Upload receiver code.

Open Serial Monitor:

- baud rate = `9600`

You will see:

Received: Hello from Arduino

How to Achieve 2 KM Range

Use Proper Antenna

Very important.

Keep Modules High

Put antenna above ground.

Use Open Area

Fields/village/open road gives better range.

Lower Frequency

India commonly uses:

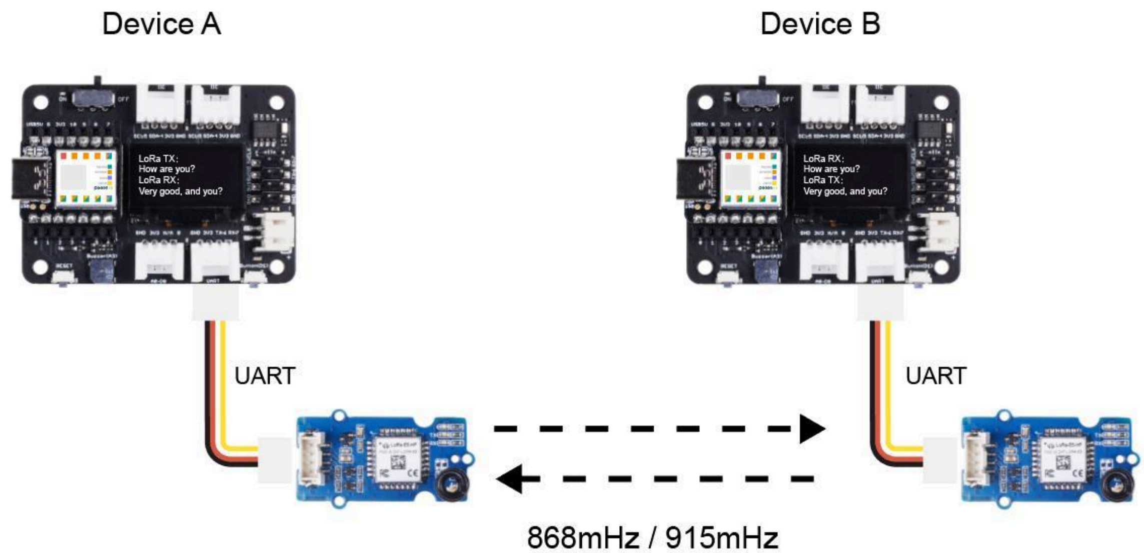
865 MHz

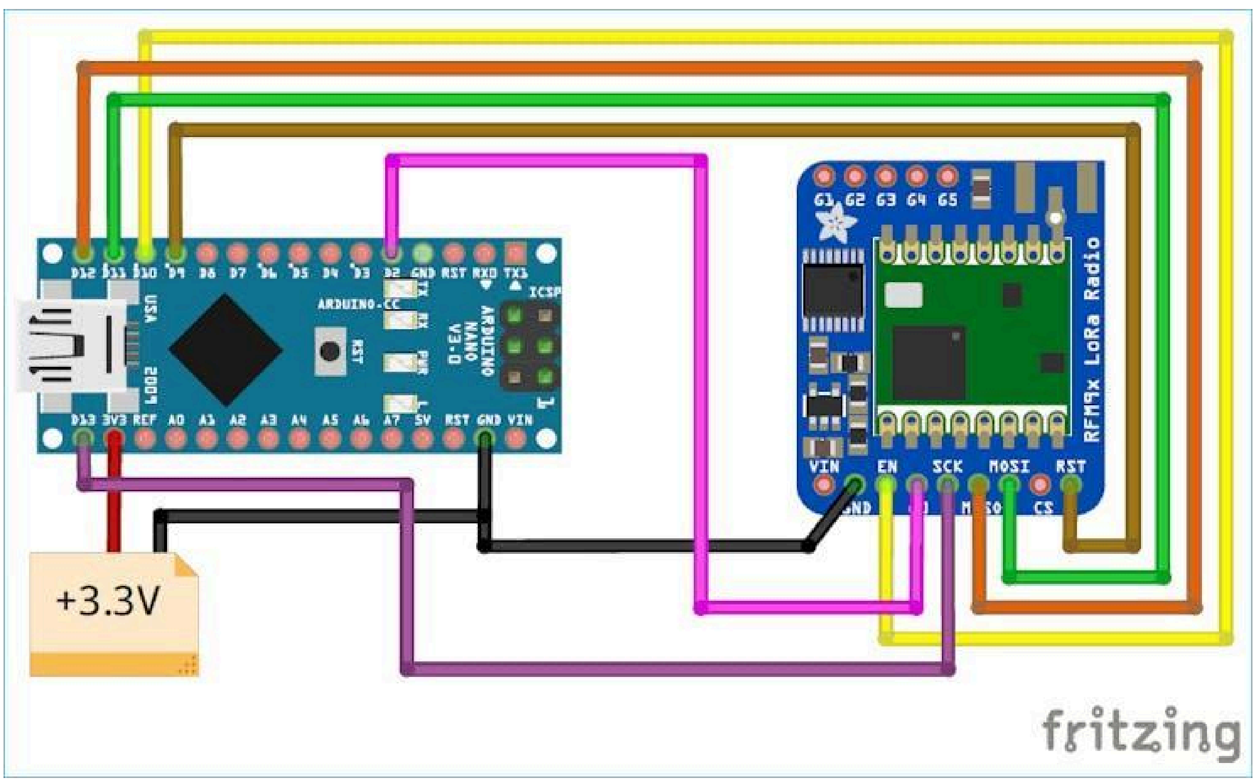
Expected Range

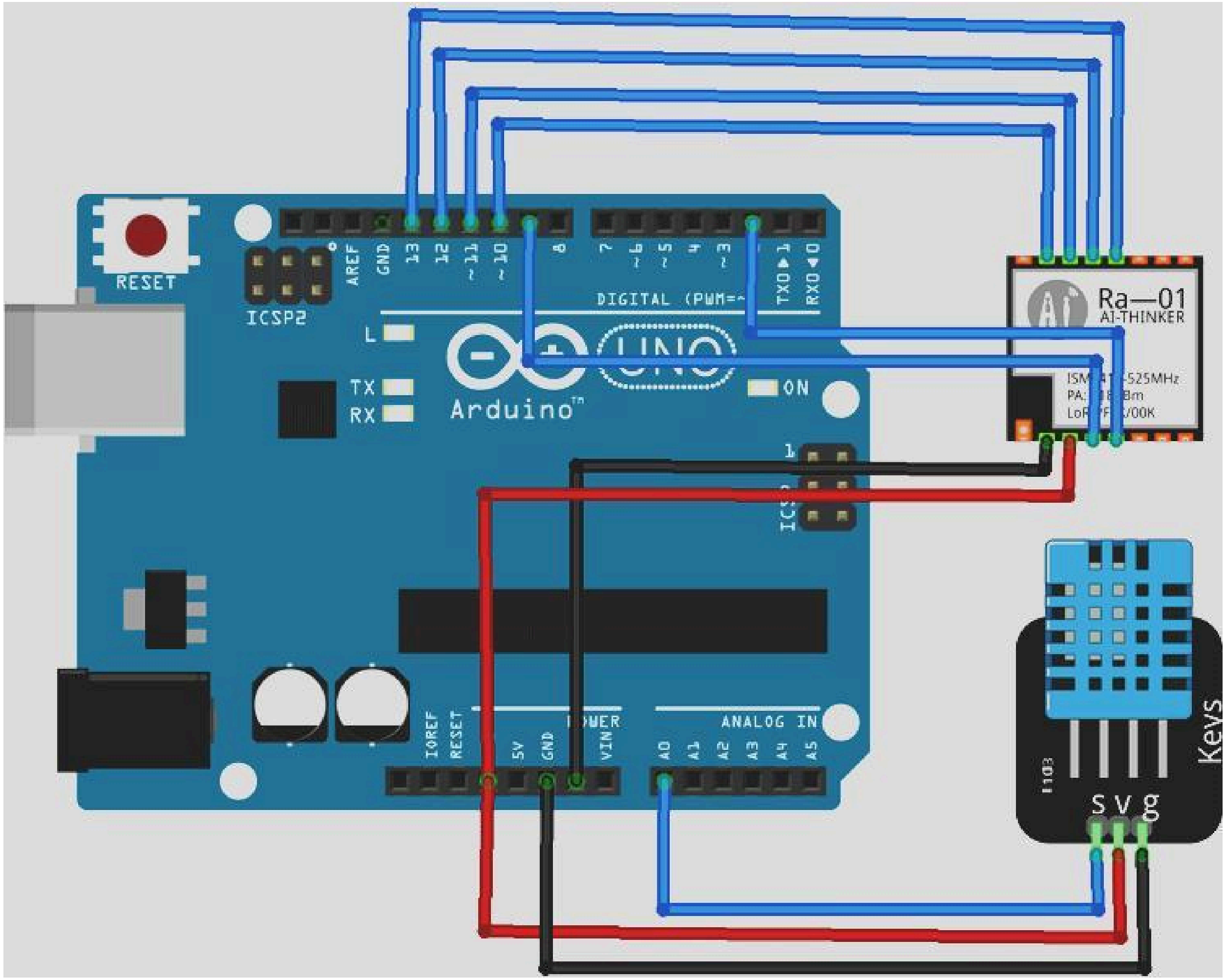
Area	Distance
City	1–3 km
Open field	5–10 km

Communication Diagram

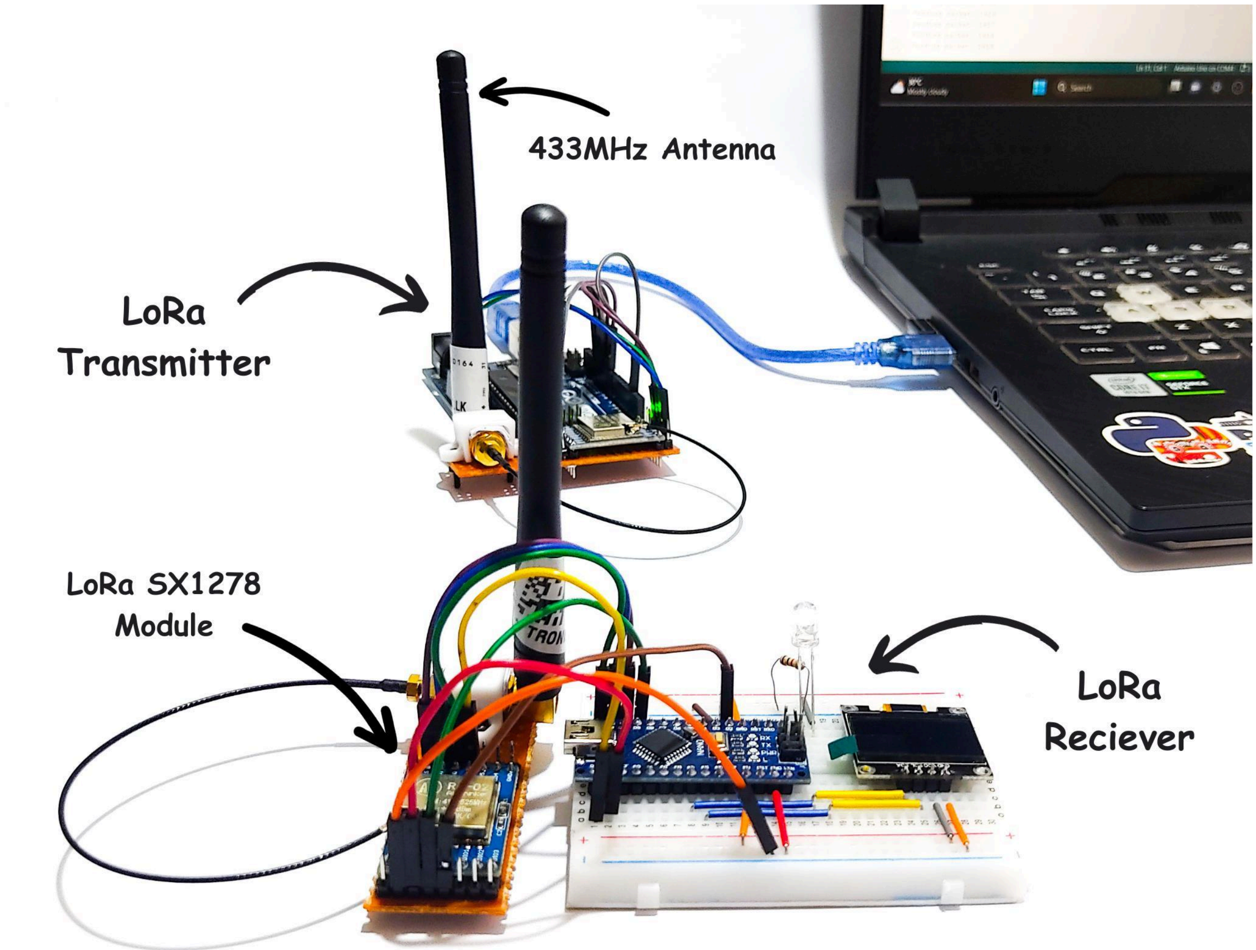
Point-to-Point Transmission with Grove - LoRa-E5







RESET
ICSP2
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX▶1
RX◀0
DIGITAL (PWM=~)
L
TX
RX
UNO
Arduino
ON
ICSP
IOREF
RESET
3V3
5V
GND
GND
VIN
POWER
ANALOG IN
A0
A1
A2
A3
A4
A5
Ra—01
AI-THINKER
Keys
s v g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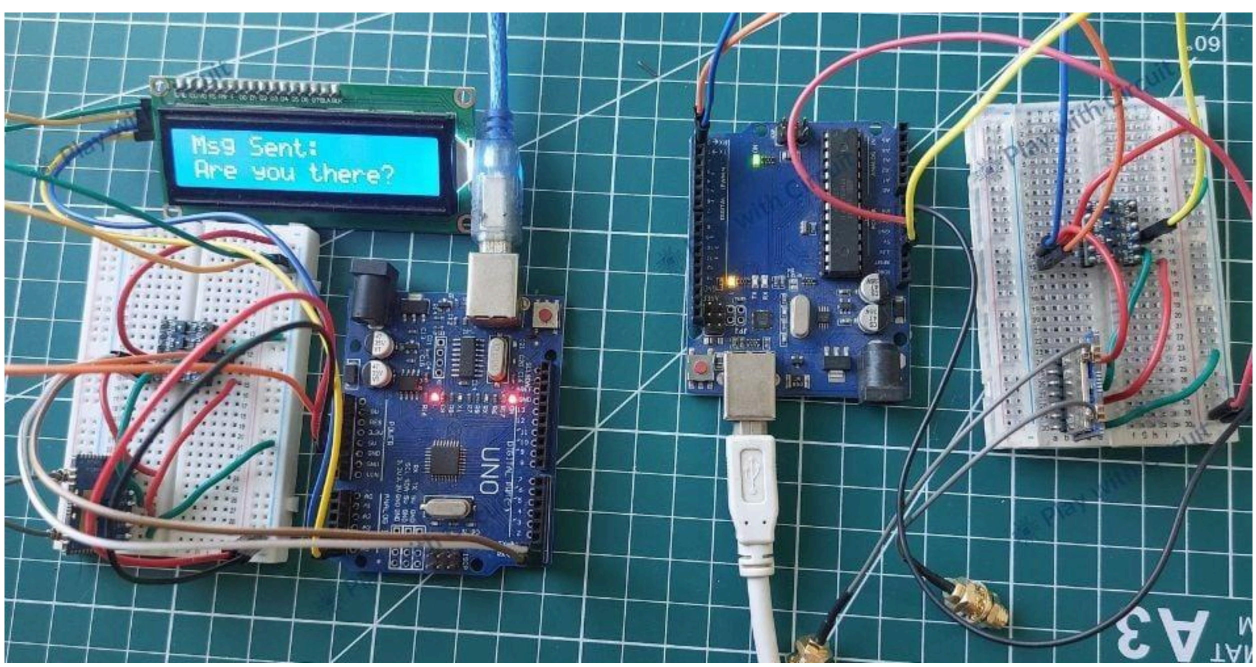
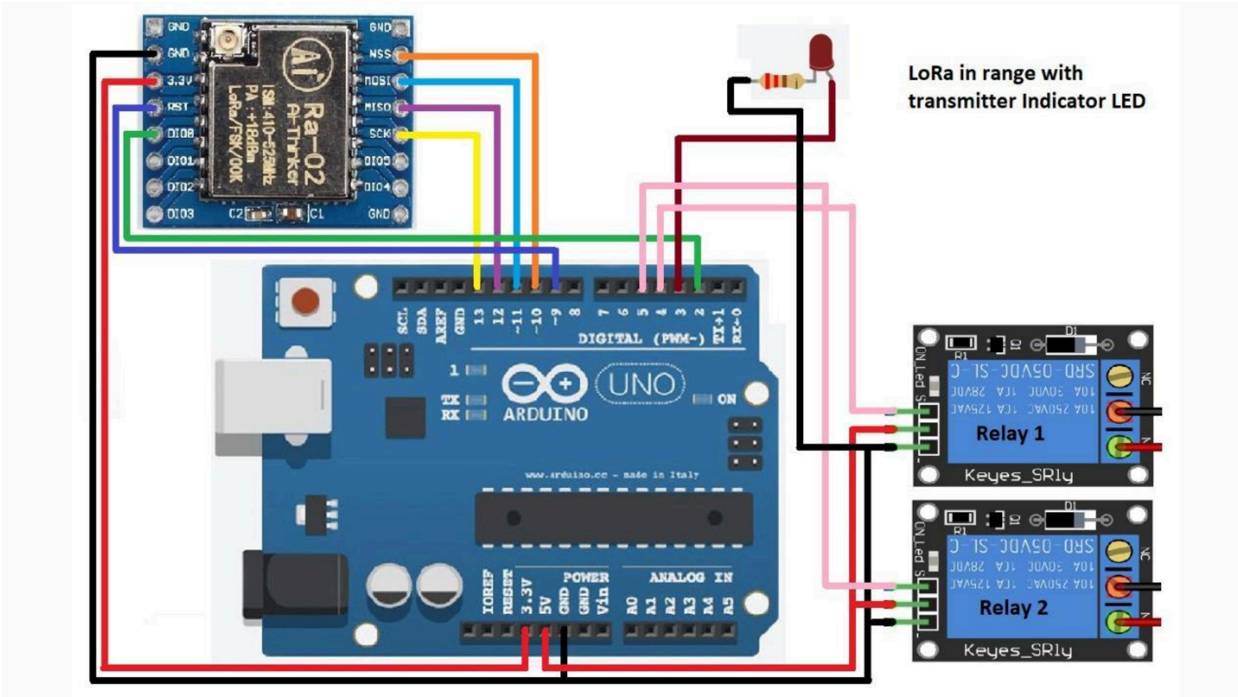
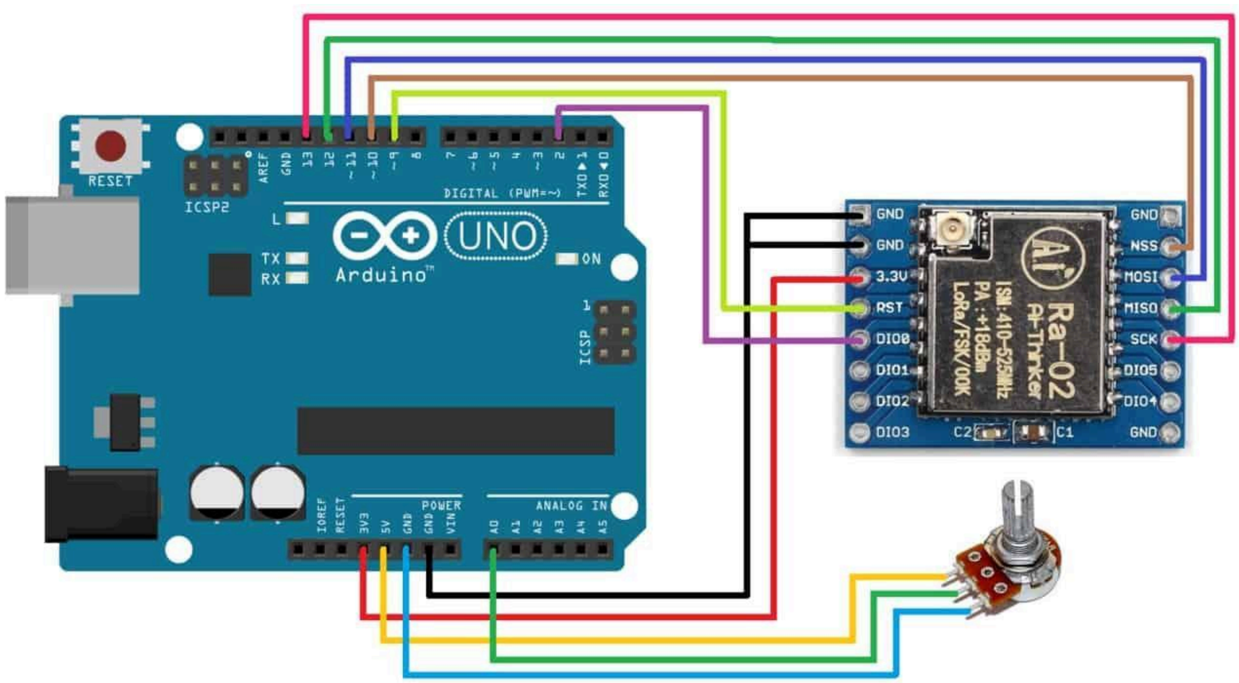

Msg Sent:
Are you there?
UNO





Arduino 1 ← LoRa → Arduino 2
Sender Receiver

Next Level Projects

You can later build:

- wireless sensor network
- GPS tracker
- drone telemetry
- chat system
- ESP32 LoRa internet gateway
- mesh network

using Arduino + LoRa.